Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / Bending
Case	5b1c4e79-1678-42e2-a64a-e589794693c5
Date	2026-11-01 18:01:16
Status	Reviewed by expert (expert)

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	92%	
dent	major	88%	
crack	critical	76%	
deformation	major	91%	

- L2: -0.11 mm
- H1: -0.05 mm

2. Root cause

Cause: Tool Wear and Die Damage

Reason: The detection of a minor scratch (32.0 mm2), major dent (64.0 mm2), and out-of-tolerance dimensions on edge L2 (-0.114 mm) and hole H1 (-0.054 mm) strongly aligns with evidence [seed-003], which cites tool wear and die damage as the root cause of scratches, dents, and dimensional variations.

Effect: Scratches, Dents, Dimensional Inaccuracy

Confidence: 68%

Supporting evidence
- **seed-003**: Tool Wear and Die Damage cause scratches, dents and dimensional variations. Preventive actions: regular tooling inspection, preventive maintenance schedule, die regrinding.

3. Solution

Corrective actions (fix this occurrence)

1. Recalibrate press brake
2. Verify material lot

Preventive actions (stop it recurring)

1. Schedule monthly calibration
2. Incoming inspection

Precautions

- Clean die surfaces prior to bending operations to prevent debris accumulation

How to verify the fix

Perform dimensional check and visual inspection on sample parts after tool regrinding

4. Other causes considered

Rank	Cause	Confidence	Reason

2	Material Inconsistency and Hardness Variation	65%	The presence of a critical crack (56.0 mm2) and major deformation (400.0 mm2) alongside dimensional deviations aligns with evidence [seed-002], which notes that material inconsistency and hardness variation lead to unpredictable deformation and cracking in sheet metal.
3	Improper Bending Force or Over Bending	62%	The detected major deformation (400.0 mm2) and dimensional deviations on edge L2 (-0.114 mm) and hole H1 (-0.054 mm) align directly with evidence [seed-001], which identifies improper bending force or over bending as the cause of deformation and dimensional inaccuracy.

5. Traceability

AI model: gemini:gemini-3.5-flash. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.